Aura: AI-Powered Real-Time Communication Coach

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***Abstract*—Effective communication is essential for success in academic, professional, and social contexts. However, access to real-time and objective feedback remains limited. *Aura* is a web-based AI application that provides live, multimodal feedback on users' speech, posture, and facial engagement. Through the integration of pretrained AI APIs such as MediaPipe and Whisper within a responsive web interface, Aura delivers instant visual and textual insights to help users improve confidence and communication skills. This paper describes the system architecture, user interface, integration pipeline, and evaluation of Aura as a human-centered AI tool for self-improvement.**

I. Introduction

Public speaking anxiety and lack of self-awareness in non-verbal behavior are common barriers to effective communication. While existing solutions focus on either speech analytics or facial expression recognition, few provide real-time, integrated feedback through a single user-friendly platform.

Aura bridges this gap by combining multimodal AI technologies with an intuitive web dashboard. The system captures live video and audio through a browser, analyzes parameters such as speaking pace, body posture, and emotion, and instantly visualizes results with actionable suggestions. Unlike earlier prototypes, the final implementation of Aura operates fully online, performing real-time inference without any separate offline model training.

II. System Architecture

Aura follows a modular architecture consisting of three main layers — Frontend Interface, Backend Processing, and Feedback Visualization.

II-A. Frontend Interface

- Built using ReactJS for a smooth and interactive user experience.
- The homepage introduces the tagline: *"Face. Train. Succeed."* and directs users to begin live training sessions.
- Captures webcam and microphone feeds directly from the browser using WebRTC.
- Displays real-time metrics: WPM (Words Per Minute), Posture (%), Eye Contact, Volume, and Emotion.

II-B. Backend Processing

- Implemented with Flask/FastAPI for asynchronous, low-latency performance.
- Uses MediaPipe for detecting facial landmarks, eye direction, and posture alignment.
- Employs OpenAI Whisper API or Google Speech-to-Text for transcription, pace, and volume analysis.
- Streams processed metrics to the frontend via WebSocket for continuous updates.

II-C. Feedback Visualization

- Aggregates live data into:
  1) **Session Dashboard:** Displays posture, eye contact, and vocal metrics in real time.
  2) **Post-Session Report:** Provides a graphical "Pacing & Confidence Trend" and textual insights.
- Generates personalized recommendations such as "Straighten your posture", "Maintain eye contact", and "Speak louder".
- The interface features smooth transitions and neon gradients for a modern design aesthetic.

TABLE I: System Performance Metrics

Metric	Target	Achieved
Average Latency	$<$ 2 seconds	1.4 seconds
Eye Contact Detection Accuracy	$>$ 80%	82.1%
Speech-to-Text Reliability	$>$ 90%	91.2%
UI Responsiveness	Smooth	Verified

III. Implementation Methodology

1) **User Flow Definition:** Defined user interactions from session start to feedback review.
2) **Web Integration:** Connected ReactJS frontend with Flask/FastAPI backend using REST and WebSocket APIs.
3) **Real-Time Processing:** Implemented frame and audio capture at 10–15 FPS for live inference.
4) **Feedback Design:** Combined visual scores and language-based advice for clarity.
5) **Evaluation:** Conducted usability testing for latency, clarity, and design responsiveness.

IV. Results and Evaluation

IV-A. User Feedback

Ten participants tested Aura through mock interviews and presentations. Key findings:

- The dashboard made performance metrics intuitive and engaging.
- Real-time feedback on posture and voice improved user self-awareness.
- Participants praised the concise interface and immediate recommendations.

V. Discussion

Aura demonstrates how web technologies can integrate AI-driven feedback into human-centered design. Its strength lies in synthesizing multimodal data into coherent, real-time insights. Limitations include sensitivity to lighting and microphone quality. Future updates may include adaptive learning, tone personalization, and multilingual support.

VI. Applications and Future Scope

- **Education:** Public speaking and communication training for students.
- **Corporate Training:** Real-time interview or presentation feedback.
- **Assistive Technology:** Confidence support for individuals with speech anxiety.

Future developments include mobile compatibility, cloud-based session analytics, and wearable integration for motion tracking.

VII. Intellectual Property and Dissemination

Aura's contribution focuses on real-time integration of visual and auditory AI insights within a cohesive web platform. While leveraging pretrained APIs, its originality lies in experience architecture and seamless feedback synthesis. The team aims to present Aura as a technical paper and explore potential IP protection under IIT Kanpur's innovation policy.

VIII. Conclusion

Aura exemplifies the convergence of artificial intelligence and user-centered design for self-improvement through immediate, actionable insights. By offering real-time communication analytics through an intuitive interface, Aura democratizes access to personalized feedback and represents the future of design-driven AI systems for behavioral development.

References

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